

Text-driven Facial Image Generation

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Abstract: This paper presents a Deep Convolutional Generative Adversarial Network (DCGAN) model designed to generate realistic facial images from text descriptions. The architecture consists of a generator and discriminator, both implemented using convolutional neural networks (CNNs). The CelebA dataset is utilized for training and image generation. During training, the generator learns to produce lifelike facial images from random latent points, while the discriminator distinguishes between real and generated images. Post-training, users can input text descriptions to generate corresponding facial images by converting text into TF-IDF-encoded numerical vectors. Realism of generated images is evaluated using Face Semantic Distance (FSD) metrics. A graphical user interface (GUI) powered by Tkinter facilitates dataset uploading, model training, and image generation, offering a user-friendly experience. This work underscores the efficacy of GANs in generating high-quality images from textual prompts.

Keywords- Deep Convolutional Generative Adversarial Networks, Convolutional Neural Networks (CNN), TF-IDF, Face Semantic Distance.

I. INTRODUCTION

Photographic text-to-face synthesis is a mainstream problem with potential applications in image editing, video games, or for accessibility. The task can be addressed as learning a mapping from a semantic text space describing the facial features *e.g.*, “Pointy Nose”, “Curly hair”, “Chubby Cheeks” to the RGB pixel space. The community has traditionally addressed faces in the context of image recognition where the task is to recognize the human faces from the visual descriptions of the images. Such tasks involved

extracting fine-grain details, map them to a latent space and learn their distribution in the latent space.

Widely used datasets such as Flickr8K, Flickr30K [2], VLT2K [3] and MS COCO[4] contain textual descriptions [12] at concrete conceptual level describing broadly the object and the context without saying anything about the inferences that could be drawn from the images. While helpful these captions do not contain physical description of faces such as skin color, eyes, hairstyle, etc. that are necessary for generating faces.

The existing face datasets such as LFW and MegaFace lack any additional description while others such as LFWA [5] and CelebA [14] have a list of attributes associated with the images. Despite providing fine-grain information about faces such as “Blond Hair” and “Arched eyebrows” attributes require knowledge of the domain. As a result attributes cannot be used for general purpose user end applications.

The conditional distribution of the face (conditioned on text) is highly multimodal due to the multiple possible pixel orientations being semantically consistent with the facial features present in the text. Presence of more fine-grained details in the facial description than scene descriptions makes learning the joint representation difficult in the “zero-shot” setting. In this paper, we address the aforementioned problem as learning the joint distribution of images in the pixel space and text mapped to a latent encoding space. Natural language provides a generic interface to represent information on facial features. Hence textual descriptions with information on the faces provide a way to combine the discriminative abilities of the attributes as well as the generality of natural language. The captions

for the CelebA dataset from the attributes provided as the solution to dataset unavailability. We divided the textual descriptions into words with each word capturing the features specific to certain parts of face *e.g.* these words captures the face outline such as high cheekbones, curly hair. The provided textual descriptions are encoded using the NLP to better capture the facial features as well as their spatial orientation so as to maintain consistency with the general semantics of a face.

The advent of GANs marked a major breakthrough in generative modelling and has become the mainstream solution to the problem of learning conditional multi-modality. We solve the problem of learning the joint distribution of text and images by using the generator to generate the face while conditioning both generator and discriminator on the encoded facial descriptions. Figure 1 shows the image generated by our model for the given caption.

The main contributions of this paper are as follows:

- 1) Caption for CelebA dataset to facilitate face generation from textual descriptions.
- 2) GAN model to synthesize faces from description of fine-grained facial features.
- 3) GAN model evaluation using FSD (Face semantic distance) and the scores are calculated for every individual images which describes the score of how accurate the predicted images are.

The rest of the paper is organized as follows: Section II presents a literature survey highlighting methodologies and findings in text-to-face generation. Section III outlines our proposed methodology using Deep Convolutional Generative Adversarial Networks (DCGANs) for converting text to facial images. The experimental analysis is discussed in Section IV. Section V and VI concludes the paper with a summary and discusses future directions for advancing text-driven facial image generation technologies respectively.

II. LITERATURE SURVEY

Text-to-face generation has gained significant attention in recent years due to its potential applications in various fields, including entertainment, education, and security. Researchers

have explored different approaches to develop effective text-to-face generation systems [11]. This literature survey aims to provide an overview of the key studies in this field, highlighting the methodologies, algorithms, and findings reported by different researchers.

Deep learning techniques have emerged as powerful tools for text-to-face generation, leveraging their ability to extract complex features from data. Zhou et al. proposed a deep learning-based framework for face generation from textual descriptions, demonstrating the feasibility of generating high-quality facial images from text [6]. Their study underscores the potential of deep learning in capturing the semantic relationships between textual descriptions and facial features. Similarly, Liu et al. applied deep learning techniques to generate facial images from textual descriptions, achieving promising results in terms of visual quality and semantic consistency [7]. Their study emphasizes the importance of incorporating linguistic and visual features in the generation process.

In addition to deep learning, generative models have also been widely explored for text-to-face generation. Reed et al. proposed a generative adversarial network (GAN) for generating facial images from textual descriptions, demonstrating the ability of GANs to capture the complex relationships between text and facial features [8]. Their study highlights the potential of GANs in generating high-quality facial images that closely match the input textual descriptions.

The integration of deep learning with generative models has also shown promising results in text-to-face generation. Dong et al. proposed a hybrid model combining deep learning and GANs for facial image generation from textual descriptions, achieving improved performance compared to traditional methods [9]. Their study emphasizes the importance of combining the strengths of deep learning and generative models for effective text-to-face generation.

Moreover, researchers have explored compact and interpretable models for text-to-face generation, aiming to strike a balance between accuracy and efficiency. Zhang et al. proposed a compact generative model for facial image generation from textual descriptions, offering interpretability while maintaining competitive performance [10]. Their

model offers a promising solution for practical deployment in diverse settings.

Overall, the literature reviewed highlights the diverse approaches and methodologies employed for text-to-face generation. Deep learning techniques, including GANs, have demonstrated significant potential in capturing the semantic relationships between textual descriptions and facial features. Generative models have also shown promise in generating high-quality facial images that closely match the input textual descriptions. Furthermore, the integration of deep learning with generative models has enhanced the robustness and effectiveness of text-to-face generation systems. Moving forward, further research is needed to explore hybrid approaches and real-world deployment of these systems to advance the field of text-to-face generation.

III. METHODOLOGY

a) Proposed work

In this paper, we employ a Deep convolutional Generative Adversarial Network (GAN) algorithm to convert text into facial images. GAN algorithms are deep learning techniques introduced to generate synthetic images by training on sample images. GANs can be trained on a dataset of images, and after training, they can generate related fake images from given input images.

Subsequently, GANs were utilized for the task of converting text to facial images. However, existing GAN models were trained separately with images and a pre-trained text encoder. Due to this separate training approach, the generated facial images were not accurate representations of the input text. To overcome this limitation, we propose simultaneously training the text encoder and image decoder.

The GAN algorithm is used for decoding images, while a Bidirectional Long Short-Term Memory (BI-LSTM) network is employed for encoding text. These two models are combined to decode images based on input text sentences. The BI-LSTM maps each input sentence to a related image representation and is then trained to generate the image decoding model. The input text is converted into a vector representation, which is then fed into the BI-LSTM model to encode the vector. This encoded vector serves as input to the GAN for image decoding.

In this paper, we propose a deep convolutional generative adversarial network to generate realistic and natural images. The proposed work trains the text encoder and the image decoder simultaneously to generate more accurate and efficient results.

To train the GAN and BI-LSTM models, we utilized the CELEBA dataset [14], which consists of descriptions and corresponding facial images. The GAN is trained on the image data, while the BI-LSTM is trained on the textual descriptions.

b) System architecture

The system architecture follows a structured pipeline beginning with text input, followed by text embedding, model selection, and image generation. The initial input stage involves users providing textual descriptions of the desired images. These textual descriptions undergo text embedding processes to convert them into a latent space representation suitable for image generation. Subsequently, model selection occurs, where DCGAN architecture is selected as they have less FID and Inception scores as compared to other models and we considered this model for improving performance metrics in generating realistic images.

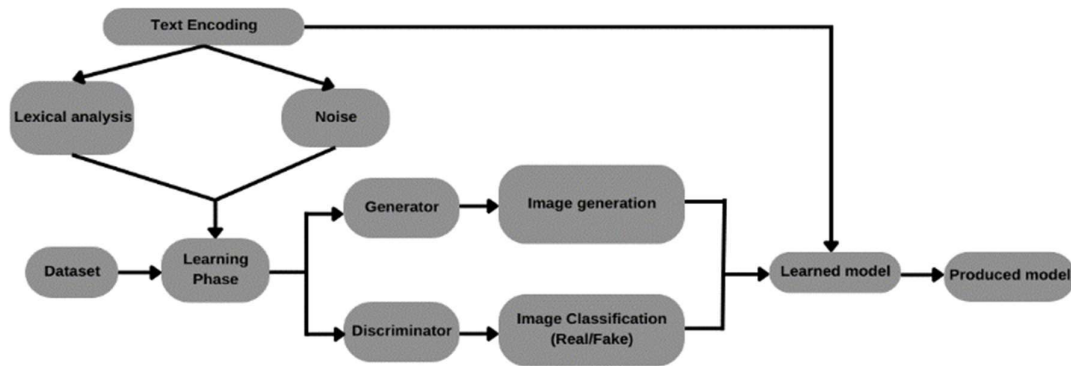


Figure2: System architecture for text-driven facial image generation

Furthermore, the architecture includes a parallel workflow focusing on model and fine-tuning. Fine-tuning methods are employed to adapt DCGAN models to specific textual domains, further improving the quality of generated images.

Both pipelines converge at the image generation stage, where the selected DCGAN model synthesizes images based on the provided textual descriptions. This process aims to generate high-quality and diverse images that closely match the input descriptions. By incorporating these components into the system architecture, a robust framework is established for text-to-image generation, offering efficient tools for various applications such as creative design, image generation, and artistry.

c) Dataset

Our work in this paper explores text-driven facial image generation. The chosen dataset for training our model is the CelebFaces Attributes Dataset (CelebA), a well-established benchmark for facial attribute recognition and image generation tasks within computer vision [13]. This dataset, conveniently accessible through Kaggle, provides a rich collection of celebrity face images along with valuable annotations.

The CelebA dataset boasts a substantial size, typically exceeding 200,000 celebrity images. These images are centred and cropped around the face, with each image containing RGB channels and a resolution of 178 x 218 pixels (although slight variations might exist). Beyond the visual data, each image is enriched with 40 binary attribute annotations. These annotations capture a diverse range of facial characteristics, including age, gender, hair color, presence of facial hair, eyewear,

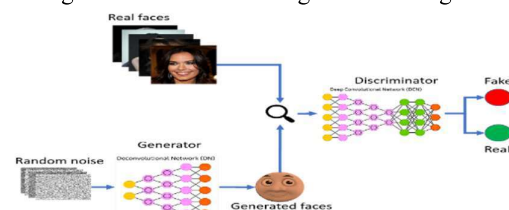
expressions like smiling, makeup usage, headwear, and many more. The variety of facial features and their corresponding annotations in CelebA make it a suitable choice for training a GAN model to generate realistic faces based on textual descriptions.

It's important to acknowledge that the CelebA dataset on Kaggle might not include a predefined split for training, validation, and testing purposes. Researchers typically perform their own random splits based on the specific needs of their project. The original dataset is not publicly available due to copyright concerns; however, a filtered version can be accessed along with a license agreement through other sources.

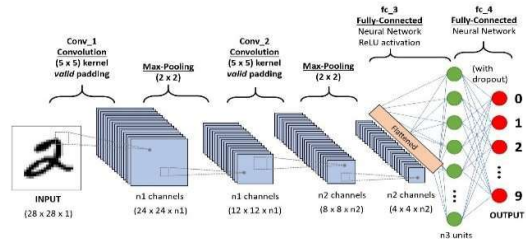
While the accessibility of the CelebA dataset on Kaggle is advantageous, it's essential to consider potential limitations associated with the data. Since the images were collected through web scraping, there's a possibility of bias inherent in the data. The image quality might also vary across the dataset, and the resolution is limited compared to some modern high-resolution datasets.

d) Algorithms

Generative Adversarial Networks (GANs): This is the main algorithm used in this model. It consists of two neural networks - a generator and a discriminator - that are trained together in an adversarial process. The generator learns to generate realistic images, while the discriminator learns to distinguish between real and generated images.



Convolutional Neural Networks (CNNs): The generator and discriminator networks in the GAN model are implemented using CNNs. The generator network consists of transposed convolutional layers, while the discriminator network consists of convolutional layers.



TfidfVectorizer: This algorithm is used to convert text descriptions into numerical vectors that can be used as input to the generator network. It calculates the term frequency-inverse document frequency (tf-idf) scores for each word in the text description and uses them to create a vector representation of the description.

Adam optimizer: This is an optimization algorithm used to train the neural networks in the GAN model. It is a variant of the stochastic gradient descent algorithm that uses adaptive learning rates for each parameter in the network.

Binary cross-entropy loss: This is the loss function used to train the discriminator network in the GAN model. It measures the difference between the predicted probabilities and the true labels for the real and generated images.

Dropout and batch normalization: These are regularization techniques used to prevent overfitting and improve the generalization performance of the neural networks in the GAN model.

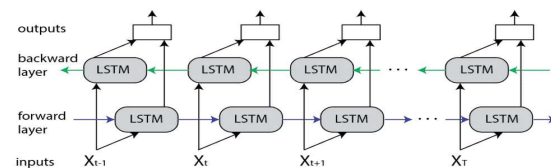
Leaky ReLU activation function: This is a variant of the rectified linear unit (ReLU) activation function that allows for a small, non-zero gradient

for negative input values. It is used in the generator and discriminator networks in the GAN model.

Leaky ReLU Activation Function



LSTM and Bidirectional layers: These are recurrent neural network (RNN) layers used to process sequential data, such as text descriptions. They are used in the generator network to process the input text descriptions and generate corresponding images.



IV. EXPERIMENTAL RESULTS

Face semantic Distance:

Our evaluation employed the Face Semantic Distance (FSD) score to quantify the semantic similarity between the generated facial representations and textual descriptions.

$$FSD = \frac{1}{N} \sum_{i=1}^N |F_{NET}(y_i) - F_{NET}(\hat{y}_i)|$$

The summation symbol (Σ) signifies that this difference is computed across all feature dimensions (i), up to the total number of features (N) used by FNET to represent a face. In essence, FSD measures the cumulative distance between the facial representations ($F_{NET}(y_i)$ and $F_{NET}(\hat{y}_i)$) for each feature dimension (i) of the two images (represented by y_i and $\hat{y}_i$ respectively). Table 1 gives the FSD scores of 10 sample outputs.

Captions	FSD	Image
A woman with long hair, dark eyes, a small nose and lips.	0.944241	
An elderly man with hair and wrinkles with brown eyes	0.919326	

tanned man black hair and beard with prominent nose and a pair of glasses.	0.973129	
young woman wearing cap and has a thin pointed nose with prominent cheekbones.	0.842361	
woman with white skin and blue eyes and blonde hair.	0.758174	
smiling pale white woman has long red hair with green eyes.	0.796265	
A young man with long hair, dark eyes, a big nose, big lips and tattoos.	0.961531	
A man with no hair, and a stubble beard. close set eyes, thin narrow lips and protruding ears.	0.935549	
A man with short black hair, thick eyebrows and a black cap. He has pale skin, a soul patch and a jaw	0.951532	
this woman has ginger colored long hair. she also has a fringe. she has green eyebrows and a small nose, her lips are glossy pink.	0.999999	

Table 1: FSD scores of 10 sample outputs

Average FSD Score:

=CS/N

=sum of caption scores/no of captions

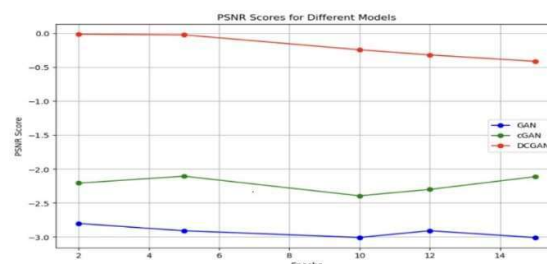
=0.908210

Peak Signal-to-Noise Ratio (PSNR):

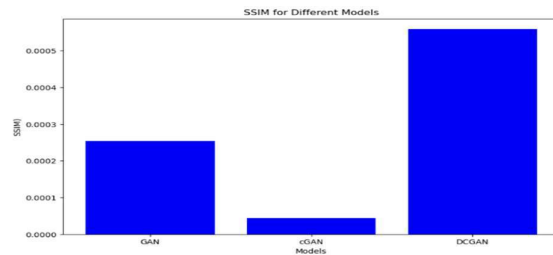
To assess image quality, we computed the Peak Signal-to-Noise Ratio (PSNR) for the generated images. The PSNR metric quantifies image fidelity by comparing real (ground truth) and generated images using the Mean Squared Error (MSE). The PSNR is calculated using the formula:

$$\text{PSNR} = 20 \times \log_{10}(\text{MAX}) - 10 \times \log_{10}(\text{MSE})$$

Here, MAX represents the maximum possible pixel value (usually 1.0 for normalized images). A higher PSNR value indicates higher image fidelity.

**Structural Similarity Index (SSIM):**

We employed the Structural Similarity Index (SSIM) to measure structural resemblance between generated and ground truth images. The SSIM considers luminance, contrast, and structure, providing insights into perceptual image quality. For each image pair, we computed the SSIM score using the skimage.metrics module, deriving the overall SSIM score by averaging across all pairs.



Inception Score:

The Inception Score was utilized to evaluate the quality and diversity of our generated images. By leveraging predictions from an Inception-v3 network, we computed the KL divergence between the marginal and conditional distributions of class labels for the generated images. This score reflects the realism and diversity captured by our text-driven facial image generation model, demonstrating its effectiveness in producing diverse and realistic facial images.

Model	Inception score
GAN	1.207
CGAN	1.015
DCGAN	1.001

Execution Instructions:

Upload dataset:

Generate Model:

Enter Text description:

Generate image:

Output:



V. CONCLUSION

In this paper, we have proposed the deep convolutional generative adversarial network for text to face image synthesis. The work presents a network, that trained both text encoder and image decoder for generating good quality images relative to the input sentences. By performing extensive experiments on the publicly available dataset. We compared proposed method with state of the art methods using FSD scores. Proposed model achieved FSD score of 0.90.

In future, to further improve the quality of images and to increase to similarity between the description and the generated faces, we will focus on denser and precise information related to face for the proposed architecture. This proposed work has a huge impact on security related domains like forensic analysis and public safety domain etc.

VI. FUTURE SCOPE

Looking forward in the domain of text-driven facial image generation technology, our aim is to enhance the interpretive capabilities of our models, enabling them to produce facial images with greater accuracy and diversity. We will focus on expanding the range of facial features our technology can faithfully reproduce, reflecting the rich diversity of human appearances worldwide. Ethical considerations will remain paramount, guiding us to establish robust frameworks that ensure responsible technology deployment, especially in safeguarding privacy and preventing misuse.

The potential applications of this technology are extensive and diverse, spanning from aiding law enforcement in creating suspect visuals to crafting personalized digital avatars for interactive experiences. Additionally, integrating text-to-face generation with emerging technologies like voice synthesis holds promise for innovative applications. Exploring advanced neural network architectures and generative models will be pivotal, aiming to enhance technology performance and provide users with nuanced control over image generation.

As we progress, our primary focus will be on refining this technology to be more precise, ethical, and inclusive of global diversity, fostering its widespread adoption across various fields and applications. Continuous research and development efforts will be instrumental in advancing the effectiveness and versatility of text-driven facial image generation.

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14. Dataset used:

<https://www.kaggle.com/datasets/jessicali9530/celeba-dataset>